Searching the Hindi Mental Lexicon: A Verbal Fluency and Spatial Arrangement Study
Report 1

Team Name: Tofu

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code: `https://github.com/krish-ag087/BRSMproject_TeamTofu.git`

1. INTRODUCTION

The mental lexicon refers to the internal organisation of words in the human mind, where words are linked by meaning and association rather than stored as an unordered list. Verbal fluency tasks provide a window into this structure by measuring the order and timing of word retrieval. When participants generate words within a semantic category, they tend to produce related words in clusters before switching to a new group (Troyer et al., 1997). The time between consecutive words, the inter-response time (IRT), reflects the cognitive processes involved in navigating semantic memory: short IRTs indicate easy within-cluster retrieval, while longer IRTs signal a switch to a new cluster (Collins & Loftus, 1975).

This study examines the structure of the Hindi mental lexicon using two complementary tasks: the Verbal Fluency Task (VFT) and the Spatial Arrangement Method (SpAM). The VFT captures the sequence and timing of word retrieval, while SpAM captures participants' explicit judgements of semantic similarity through a drag-and-drop arrangement task. Data were collected from 35 Hindi-speaking participants across four semantic domains: Animals, Foods, Body Parts, and Colours. A notable challenge in this dataset is script mixing. Approximately 30% of responses were typed in Roman script (e.g.,

"kutta") rather than Devanagari (e.g., "कुत्ता"). This affects frequency analyses and is addressed as a key limitation throughout.

2. METHODS

2.1 Participants

The dataset contains responses from 35 participants aged 19–27 (mean age 23.1, SD 2.0). Participants self-reported demographic information and Hindi language proficiency on a 1–5 scale (Hindi reading: M=4.20, SD=0.82; Hindi writing: M=3.49, SD=0.94). Not all participants completed all domains: Animals (N=35), Foods (N=35), Body Parts (N=24), Colours (N=11).

2.2 Tasks

The experiment consisted of two tasks. In the Verbal Fluency Task (VFT), participants typed as many words as possible within a semantic category within three minutes, pressing Enter after each word. The software recorded each word and the inter-response time (IRT) between responses. In the Spatial Arrangement Method (SpAM), participants arranged the words they had generated onto a shared canvas according to semantic similarity. Words placed closer together were interpreted as more semantically related. Normalised (x, y) coordinates (range 0–1) were recorded for each word's final position. The Furniture practice block always came first. The remaining domains (Animals, Foods, Body Parts, Colours) were presented in counterbalanced order across participants, with each participant completing 3 domains. This explains the unequal N per domain: Animals (N=35), Foods (N=35), Body Parts (N=24), Colours (N=11). The exit survey followed the final domain.

2.3 Analysis

All analyses were conducted in Python using numpy, scipy, sklearn, and matplotlib.

Descriptive statistics: Word counts and IRTs were summarised per domain (mean, SD, range). IRT curves were plotted by word position to examine retrieval dynamics across the word sequence.

H1 — Semantic Clustering: K-means clustering (k=4) was applied to each participant's SpAM coordinates to identify semantic clusters. Consecutive word pairs in the VFT sequence were labelled as within-cluster or between-cluster based on cluster membership. An independent samples t-test compared within-cluster vs between-cluster IRTs per domain.

H2 — SpAM Distance vs IRT: For each word appearing in $\geq$3 participants' SpAM

data, the mean normalised position was calculated. Mean Euclidean distance to the 3 nearest neighbours in SpAM space was computed as a measure of semantic neighbourhood density. Pearson correlation tested the relationship between neighbourhood distance and mean IRT per word.

H3 — Proficiency vs Retrieval: Pearson correlations tested the relationships between self-rated Hindi reading and writing proficiency and (a) total words produced and (b) mean IRT.

Multiple Comparisons: With 4 domains × 3 hypotheses = 12 tests, Bonferroni correction was applied: $\alpha = 0.05/12 = 0.0042$. Results significant at $p<0.05$ but above 0.0042 are reported as borderline.

3. RESULTS

3.1 Descriptive Statistics

Table 1: Word count and IRT summary per domain

Domain	N	Mean Words	SD	First Word IRT	Subsequent IRT
Animals	35	10.4	4.5	10.64s	4.85s
Foods	35	9.3	4.0	15.30s	4.94s
Body Parts	24	8.6	3.0	13.09s	5.46s
Colours	11	13.4	3.3	6.45s	3.72s

Animals produced the highest mean word count (M=10.4), while Body Parts produced the fewest (M=8.6). Colours showed the highest mean but with only N=11, this result should be interpreted with caution. First-word IRTs were consistently 2–4 times longer than subsequent IRTs across all domains, reflecting an initial broad memory search before faster cluster-based retrieval begins.

IRT dropped sharply from the first word to the second and then stabilised, consistent with the idea that the first word requires a broader memory search while subsequent words are retrieved within already-located semantic clusters.

3.2 Most Frequently Produced Words

The most frequently produced words reflected culturally salient items. For Animals, कुत्ता/kutta (dog) and शेर (lion) were most common. For Foods, चावल (rice) was most frequent (14 participants). For Body Parts, कान (ear) was most frequent (14 participants), followed by नाक (nose, 13) and हाथ (hand, 13).

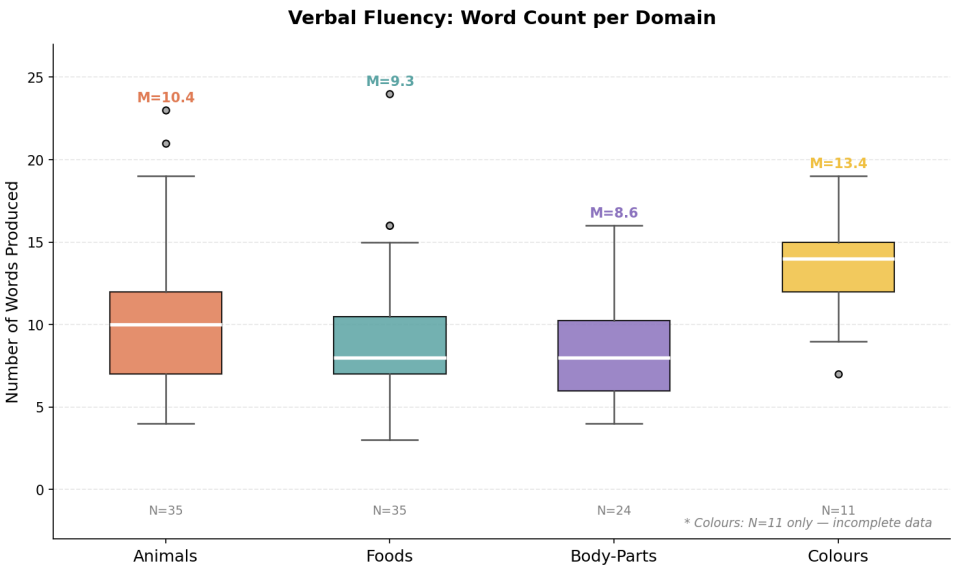

Figure 1: Boxplots of word counts per domain.

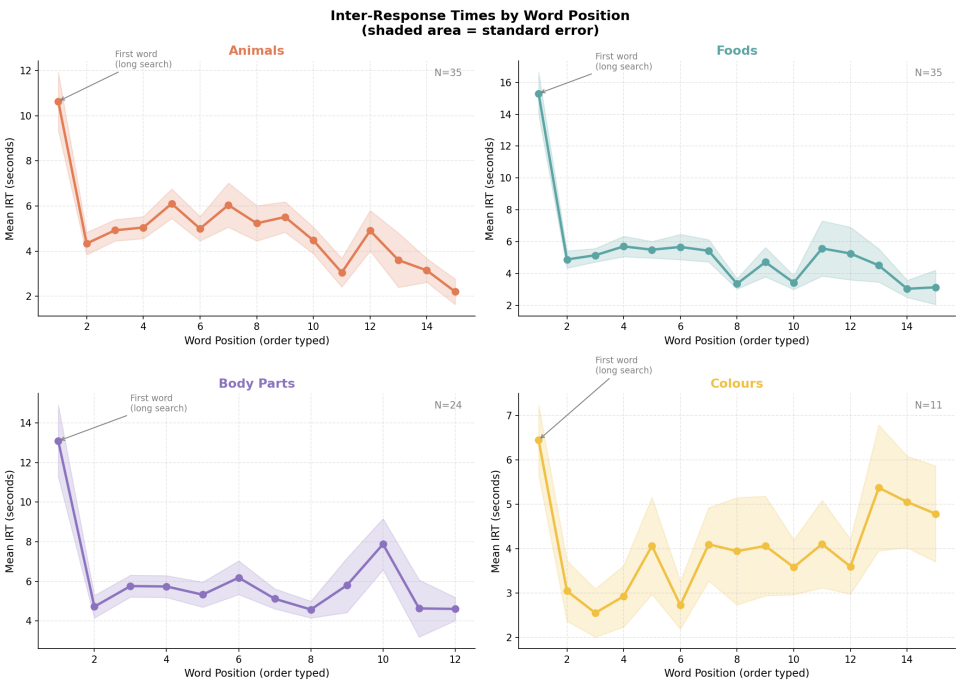

Figure 2: Mean IRT by word position across all four domains. Shaded regions show ±1 standard error. N decreases at later positions as fewer participants reached those positions.

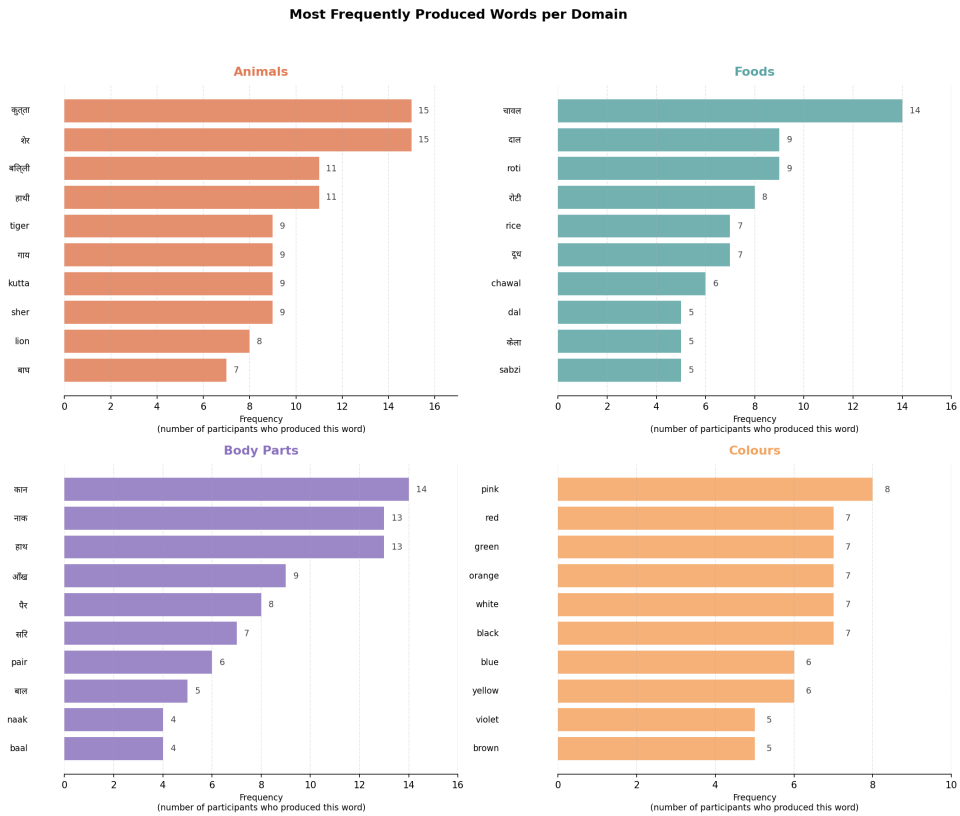

Figure 3: Top 10 most frequently produced words per domain. Script variants of the same word (e.g., chawal/चावल) appear as separate entries due to unnormalised data.

3.3 H1 — Semantic Clustering

Table 2: H1 independent samples t-test results

Bonferroni corrected threshold: $\alpha = 0.0042$

Animals and Foods showed within-cluster IRTs approximately 1 second shorter than between-cluster IRTs, significant at $p<0.05$ but above the Bonferroni threshold. Body Parts showed no clustering effect — within-cluster IRTs were marginally longer than between-cluster IRTs, suggesting body parts are retrieved as a sequential list rather than semantic clusters. Colours was underpowered (N=11) and showed a non-significant trend in the correct direction.

3.4 H2 — SpAM Neighbourhood Distance Predicts IRT

Table 3: H2 Pearson correlation results

Bonferroni corrected threshold: $\alpha = 0.0042$

A strong significant positive correlation was found for Body Parts ($r=0.672$, $p=0.0016$) — words placed further from their neighbours on the SpAM map took longer to retrieve in the VFT. Colours showed a borderline correlation ($r=0.589$, $p=0.027$). Animals and Foods showed no significant relationship.

Domain	Within IRT	Between IRT	Difference	t	p
Significant?					
Animals	4.24s	5.32s	+1.08s	-2.771	0.0059
∼ borderline					
Foods	4.26s	5.38s	+1.11s	-2.709	0.0072
∼ borderline					
Body Parts	5.57s	5.24s	-0.33s	0.658	0.5118
× Not significant					
Colours	3.26s	3.97s	+0.71s	-1.380	0.1700
× Not significant					

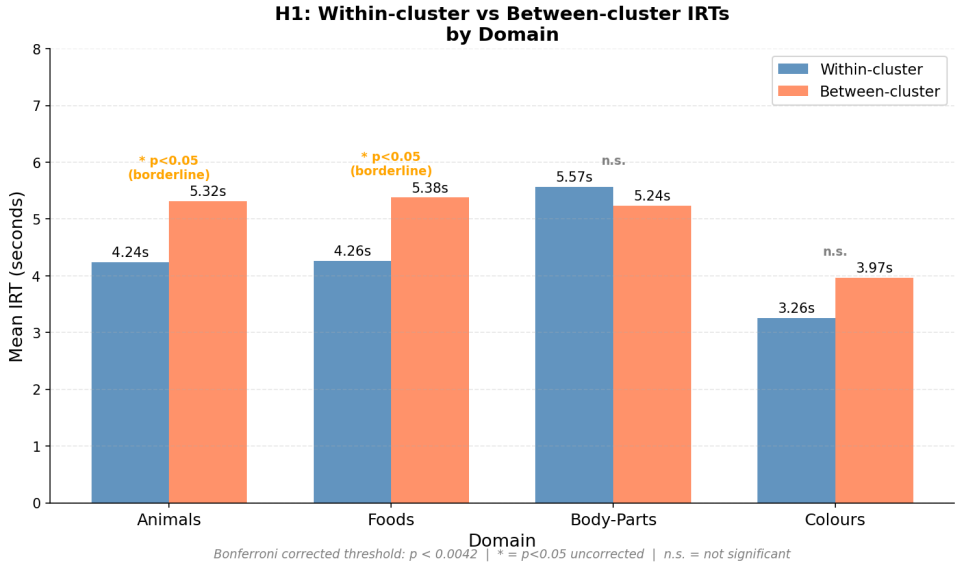

Figure 4: Within-cluster vs between-cluster mean IRTs per domain. * = p<0.05 uncorrected (borderline), n.s. = not significant. Bonferroni corrected threshold = 0.0042.

Domain	Words Analysed	r	p	Significant?
Animals	34	0.005	0.979	× Not significant
Foods	21	-0.168	0.467	× Not significant
Body Parts	19	0.672	0.0016	✓ Significant (Bonferroni)
Colours	14	0.589	0.027	∼ borderline

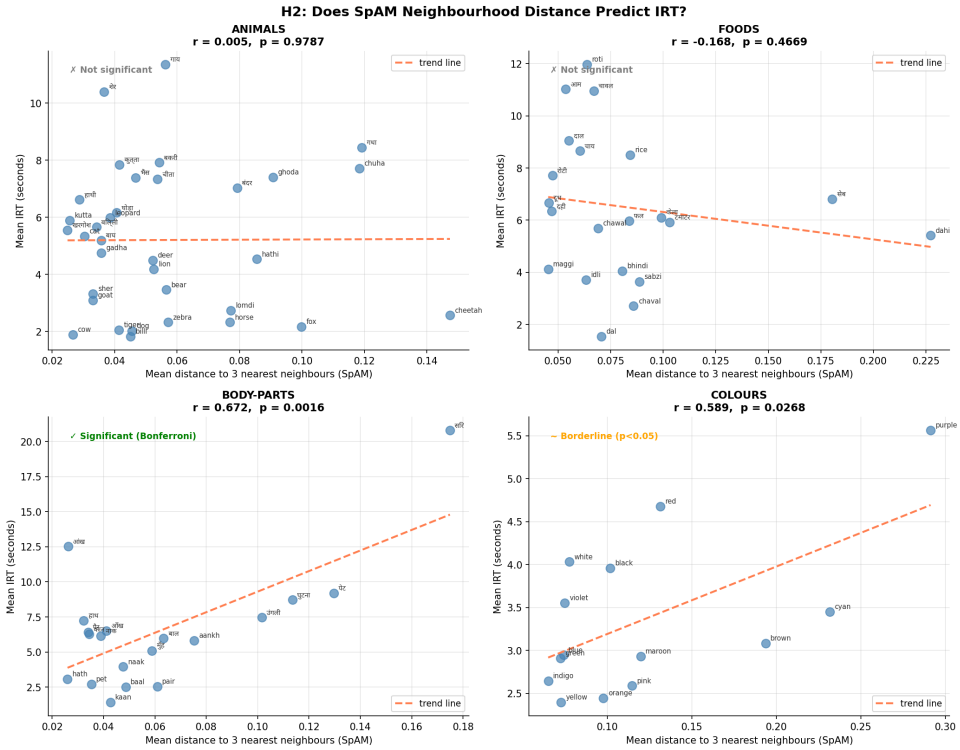

Figure 5: Scatter plots of mean SpAM neighbourhood distance (x-axis) vs mean IRT per word (y-axis) for each domain. Each dot is one word. Dashed line shows linear trend.

3.5 H3 — Hindi Proficiency Predicts Retrieval

Table 4: H3 Pearson correlation results

Test	r	p	Significant?
Hi_Read vs Total Words	-0.129	0.460	× Not significant
Hi_Write vs Total Words	-0.161	0.356	× Not significant
Hi_Read vs Mean IRT	0.271	0.116	× Not significant
Hi_Write vs Mean IRT	0.254	0.140	× Not significant

No significant correlations were found between Hindi proficiency and retrieval performance. This is likely due to insufficient variance in the sample — most participants rated themselves 4 or 5 out of 5 on the proficiency scale, leaving too little range to detect a meaningful relationship.

4. CONCLUSION AND NEXT STEPS

This study examined semantic organisation in the Hindi mental lexicon using verbal fluency and spatial arrangement tasks. Animals and Foods showed evidence of semantic clustering, with within-cluster inter-response times approximately 1 second shorter than

between-cluster transitions, though these effects did not survive Bonferroni correction. Body Parts showed a different retrieval pattern, likely reflecting a sequential body-schema representation rather than semantic clustering. SpAM neighbourhood distance significantly predicted IRT for Body Parts (r=0.672, p=0.0016), supporting a semantic network model of lexical retrieval for this domain.

A key limitation is script variation between Devanagari and Roman Hindi (∼30% Roman responses), which prevented reliable normalisation of lexical items and likely underestimates clustering strength. The absence of a proficiency effect may also reflect insufficient variance in the sample — most participants rated themselves 4 or 5 out of 5, making it difficult to detect a real relationship. Future work will focus on: (1) build a comprehensive normalisation dictionary merging Devanagari, Roman script, and spelling variants (using fuzzy string matching); (2) rerunning all analyses on cleaned data; and (3) applying ANOVA and regression to formally compare IRT distributions across domains

In addition, three further analyses will extend the current framework. First, we will test position effects on IRT to examine whether retrieval slows as participants progress through the fluency list, reflecting a potential lexical exhaustion effect. Second, we will investigate typicality in early retrieval, analysing whether participants tend to begin with the same high-frequency or prototypical category members and identifying the most common word produced at position one. Third, we will examine multilingual effects on lexical retrieval, testing whether participants who report knowledge of more languages produce more or fewer Hindi responses compared to Hindi-dominant speakers. Together, these analyses will provide additional insight into temporal search dynamics, prototype structure, and the influence of multilingualism on semantic retrieval.

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CONTRIBUTION

Himani: Handled loading and parsing the JSON dataset, cleaning demographic data, computing descriptive statistics (word counts and IRTs per domain), generating the word count and IRT position plots.

Krish: Implemented k-means clustering on SpAM coordinates, labelled within/between cluster word pairs, ran independent samples t-tests for H1, computed SpAM neighbourhood distances and Pearson correlations for H2, generated the corresponding plots.

Sajiv: Ran Pearson correlations for H3, applied Bonferroni correction across all 12 tests, generated the top words frequency plot.

Each member contributed equally in making the report and presentation.